

OPEN SOURCE

APACHE 2.0

AI Gateway & Registry

One Governed Front Door for Any AI Resource: Model Inference, MCP Servers, A2A Agents, Skills, and REST APIs

Now a full **AI gateway and registry**, not just an MCP gateway: since 1.30.0 the same governed front door fronts model inference, MCP servers, A2A agents, and any REST backend.

MODEL INFERENCE

MCP SERVERS

A2A AGENTS

AGENT SKILLS

CUSTOM ENTITIES

ENTERPRISE READY

GitHub Repository

AWS Workshop

Last updated: September 16, 2026

What we'll cover

01	The Problem & the Platform
02	Registry: Discover, Trust & Federate
03	Gateway: Connect Securely
04	Govern: Policy, Identity, Audit
05	Run It: Operations, Proof & Roadmap
06	Get Started

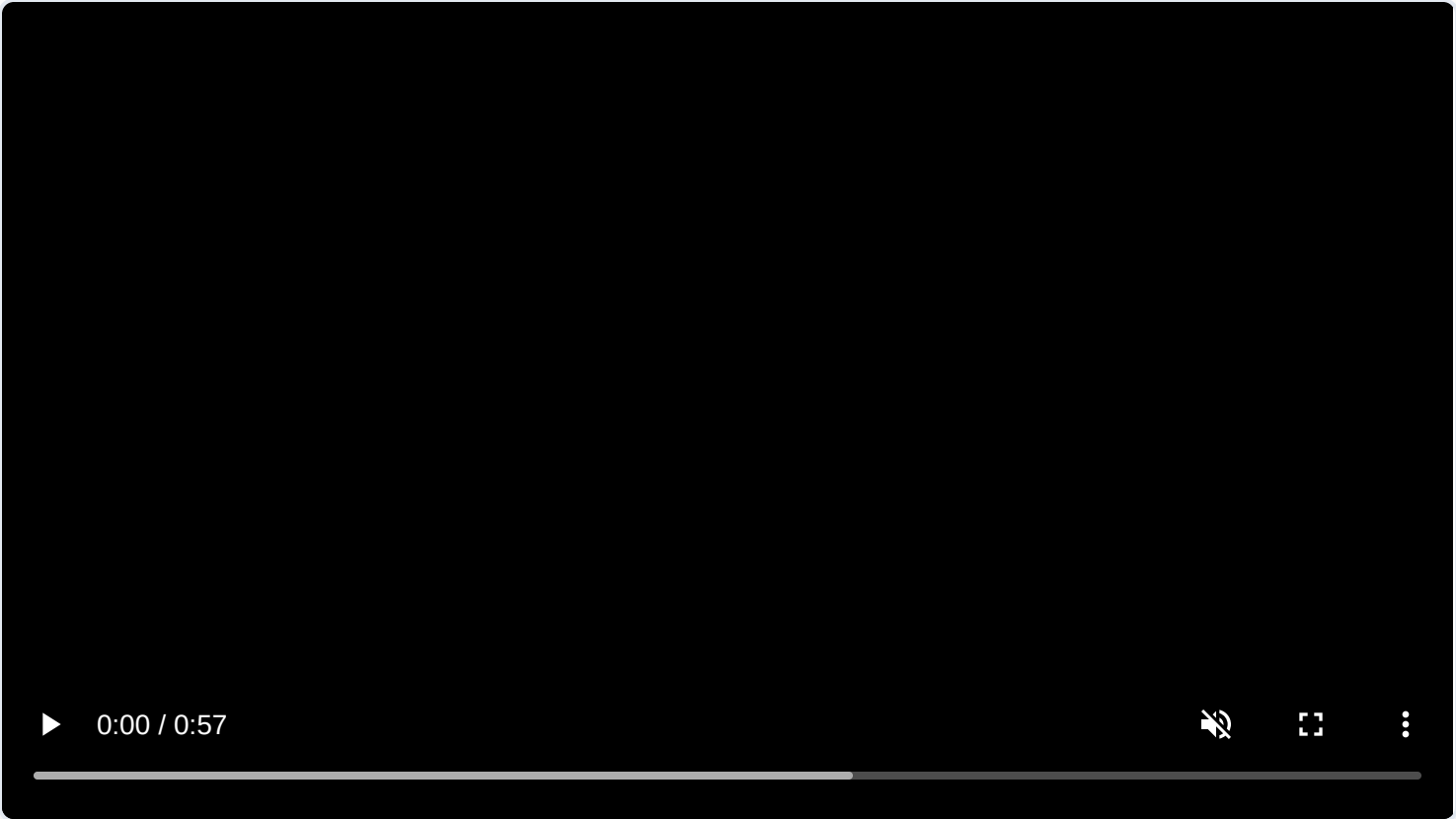
01

The Problem & the Platform

What AI asset sprawl costs you, and the unified control plane that fixes it.

Why We Built This

57 seconds: the problem, how the catalog grew, and the two halves that solve it.



Video not playing? [Watch the 57-second overview](#)

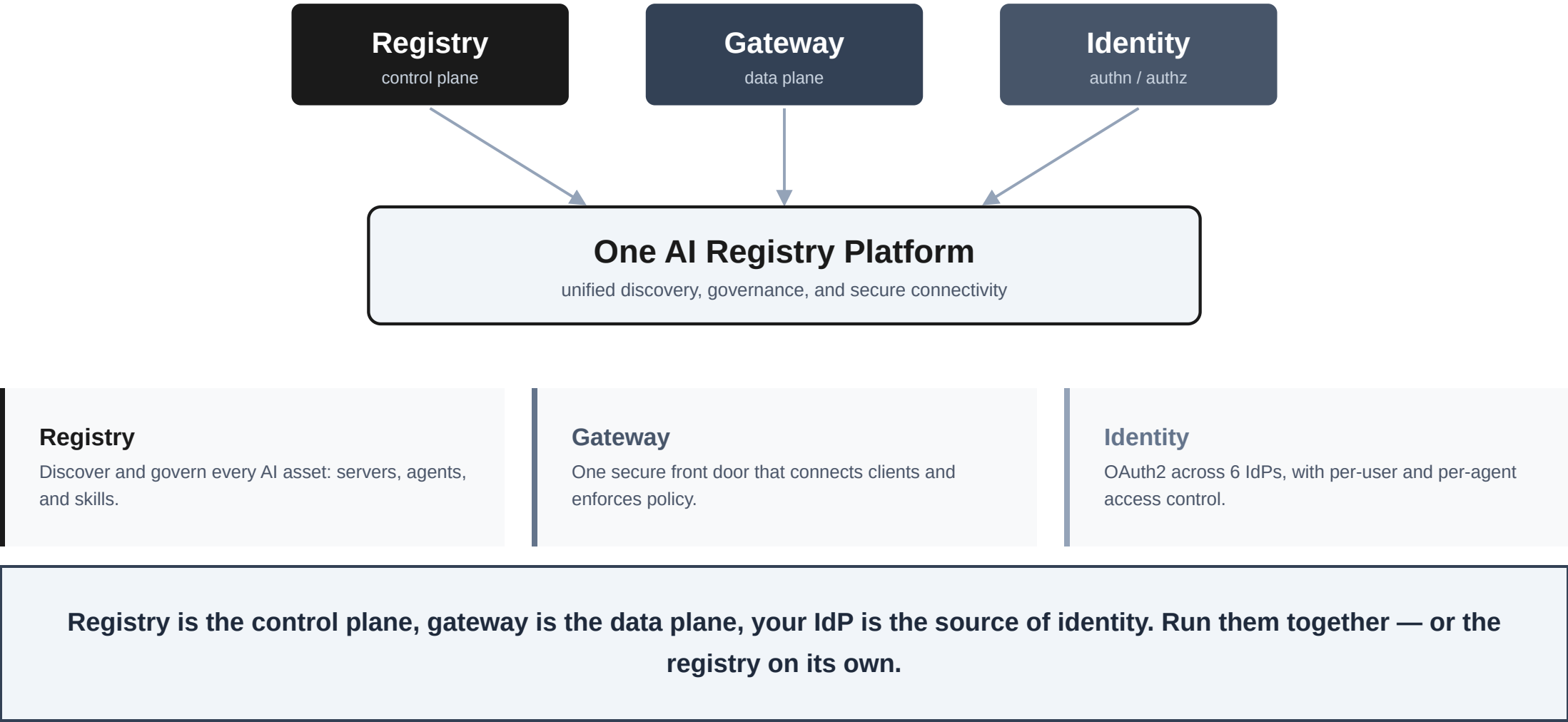
The AI Asset Sprawl Problem

Teams ship MCP servers, agents, and skills faster than anyone can track. Four needs surface again and again.

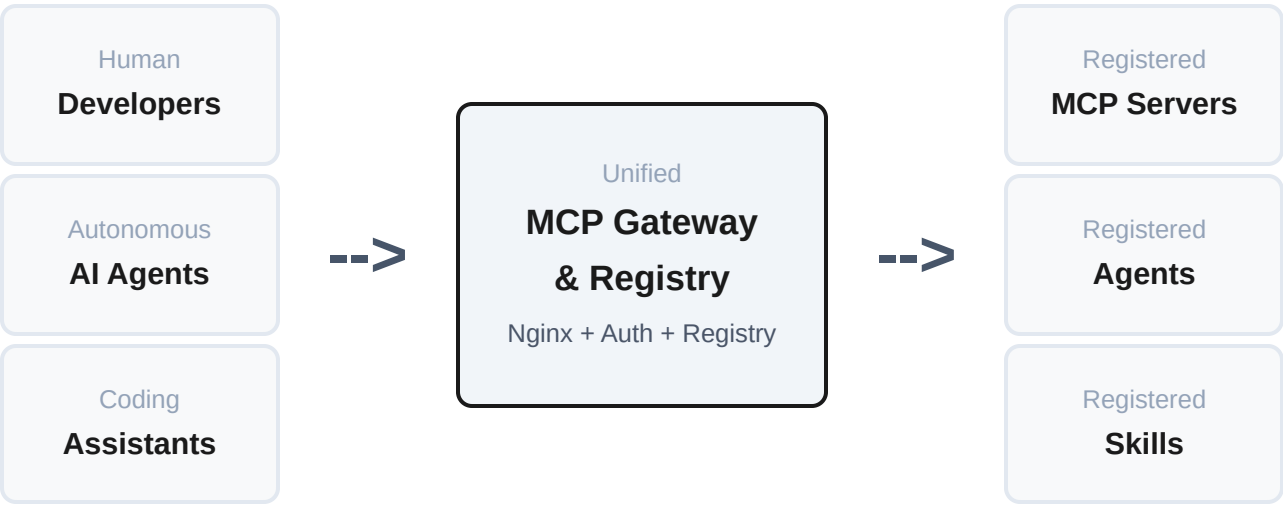
 Discover Find the right server, agent, or skill across a fast-growing catalog.	 Trust Know what you connect to is safe before it goes live.	 Connect Securely One front door, instead of every agent holding every credential.	 Govern Control access, versions, and audit across everything.
Point tools solve one need each. The real requirement is a single platform — and these four needs are the story of this deck: Discover, Trust, Connect, Govern.			

What You Really Need: One Platform

A registry, a gateway, and identity for authn and authz, working as one.



Platform Architecture



Servers and agents can be hosted **anywhere**:

- AMAZON BEDROCK AGENTCORE
- AMAZON EKS
- OTHER AWS COMPUTE
- THIRD-PARTY SAAS

Skills are sourced from **GitHub or private Git repositories** (PAT or GitHub App auth, GitHub Enterprise supported).

Single connection point with **dual authentication** for human users and machine-to-machine agent flows.

Platform Demo



Video not playing? [Watch the platform demo](#)

02

Registry: Discover, Trust & Federate

It starts with the registry: any AI asset, findable by humans, agents, and coding assistants — vetted before it goes live, federated across every registry you have.

Intelligent Discovery

Hybrid Search

- ◆ Vector similarity + keyword matching
- ◆ Unified search across servers, tools, agents, skills
- ◆ Name-boost for exact match relevance
- ◆ Tag-based and metadata filtering

Flexible Embeddings

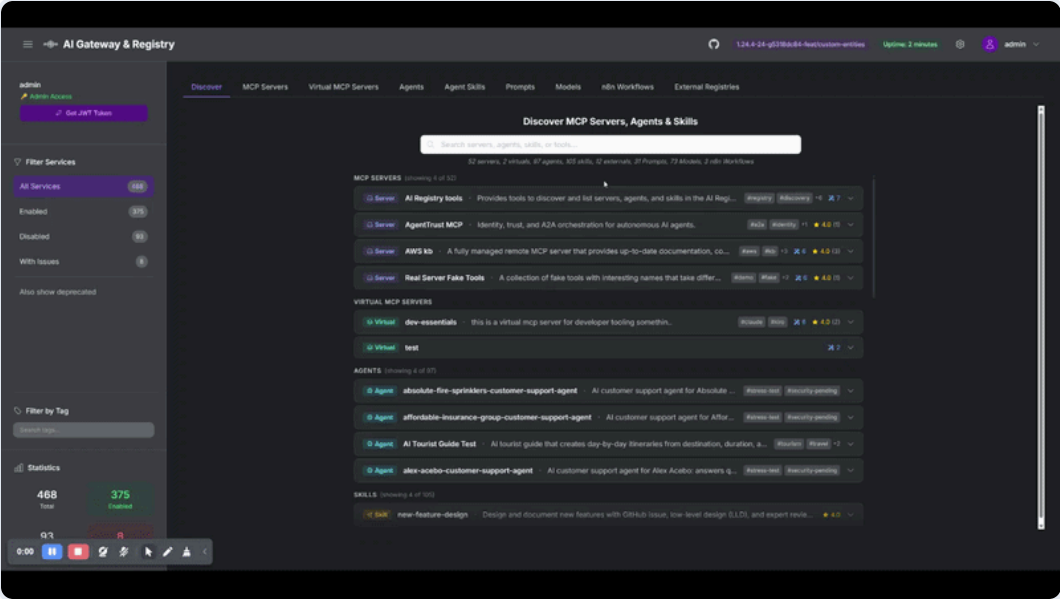
- Local sentence-transformers
- OpenAI embeddings
- Amazon Bedrock Titan
- 100+ LiteLLM-supported providers

Humans, coding assistants, and agents all use the same search. Agents discover tools *and other agents* at runtime via natural language -- **"find an agent that can book flights"** -- enabling dynamic orchestration instead of hardcoded wiring.

From MCP Registry to AI Asset Registry

It started with MCP servers, then agents and skills. Now admins define **custom entity types** with their own schema, so the same registry holds *any* AI asset your teams produce.

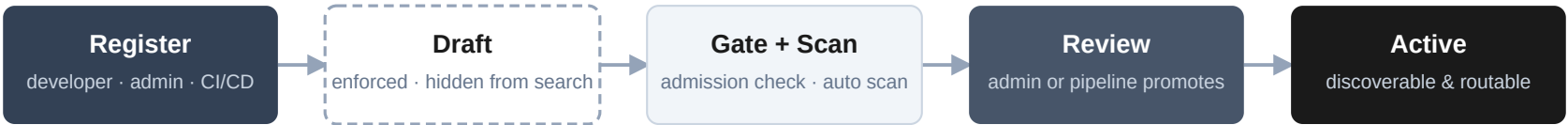
- N8N WORKFLOWS
- POLICIES
- PROMPT TEMPLATES
- MODEL CARDS
- DATASETS
- EVALUATIONS



Each custom entity type gets its own tab, schema-driven registration, semantic + lexical search, voting, and registry cards, with the same governance and audit trail as everything else.

Governed Registration: Draft → Review → Active

Developers, admins, or a CI/CD pipeline register assets self-service — but nothing goes live without review. Access control and lifecycle stages keep the flow secure and governed.



Optional webhooks notify your pipeline or admins at registration and when scans complete

Self-Service Intake

Teams and CI/CD pipelines register servers, agents, and skills themselves. Every new asset is forced to land in **draft** — registering straight to active is rejected.

Register Yes, Promote No

Scoped permissions separate the two: non-admin groups can register but lack the change-lifecycle permission, so only admins — or your pipeline — can promote to active.

Lifecycle Stages

Draft → beta → active → deprecated. Search and routing see only active, enabled assets — drafts, deprecated, and scan-flagged items stay hidden.

Assembled from built-in primitives — group scopes, one config parameter, webhooks, and the lifecycle API — so the workflow matches **your** governance process. ► [Watch the end-to-end demo](#)

Trust: Nothing Unvetted Reaches Production

Inside that workflow, every server, agent, and skill is automatically vetted — before the first call is ever routed to it.

Supply-Chain Scanning

Cisco AI Defense scanners check every MCP server, A2A agent, and skill at registration — YARA pattern matching plus optional LLM analysis for injection, XSS, and hardcoded secrets. Unsafe items are held for review.

Admission Control

An optional registration gate approves or denies assets before they are persisted — and it fails closed: if the gate is unreachable, registration is blocked, not waved through.

Verified Agent Identity

Agent Name Service (ANS) integration: DNS + PKI for agents. Owners link an ANS ID backed by an X.509 certificate, and the registry verifies it — the way SSL certificates let you trust websites.

Trust badges, visibility controls, and star ratings then tell every consumer **what is safe to use.**

Supply Chain Security: Cisco AI Defense

EXTERNAL INTEGRATION

Three open-source scanners automatically scan every MCP server, A2A agent, and Agent Skill before it reaches production.

MCP Scanner

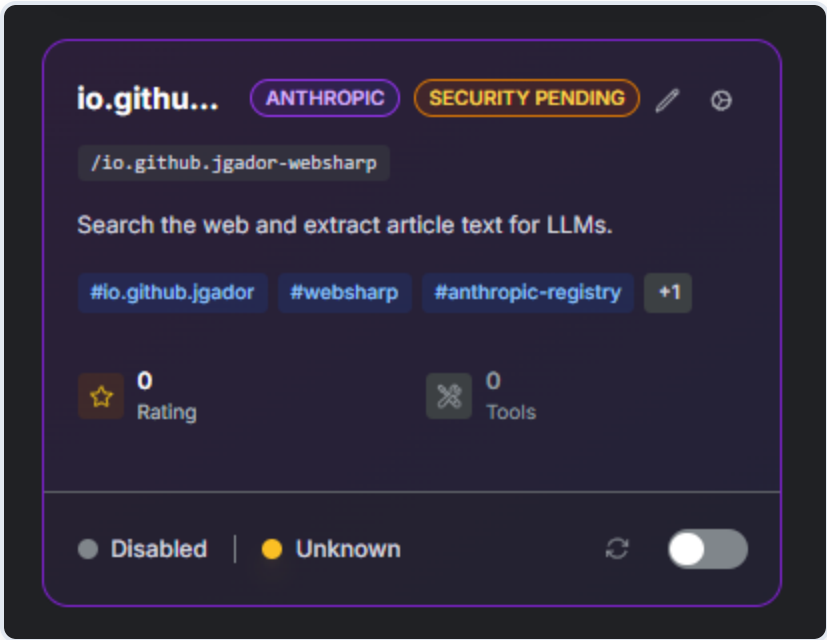
YARA pattern matching and optional LLM analysis for MCP server tool definitions. Detects injection, XSS, and hardcoded secrets.

A2A Scanner

Heuristic and spec validation for Agent-to-Agent protocol agents. Catches protocol violations and malicious behaviors.

Skill Scanner

Static and behavioral analysis of SKILL.md files. Detects prompt injection, privilege escalation, and social engineering.



Unsafe items are auto-disabled with a "security-pending" tag for admin review

Scan on registration, on-demand rescan, and periodic registry-wide audits -- powered by [cisco-ai-defense](#) open-source scanners

Agent Trust: Agent Name Service

EXTERNAL INTEGRATION

DNS + PKI for AI Agents -- just as SSL certificates let you trust websites, ANS lets you trust agents.

Structured Identity

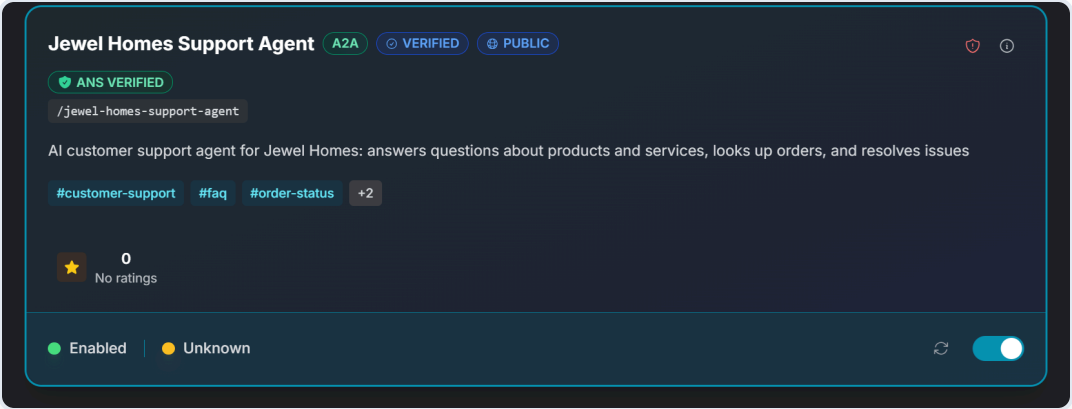
Each agent gets an ANS Name and X.509 certificate issued by a Registration Authority.

Bring Your Own ANS ID

Owners link their ANS Agent ID to registry entries. Registry verifies via GoDaddy ANS API.

Production Resilience

Circuit breaker, retry with backoff, 6-hour background re-verification, and rate limiting.



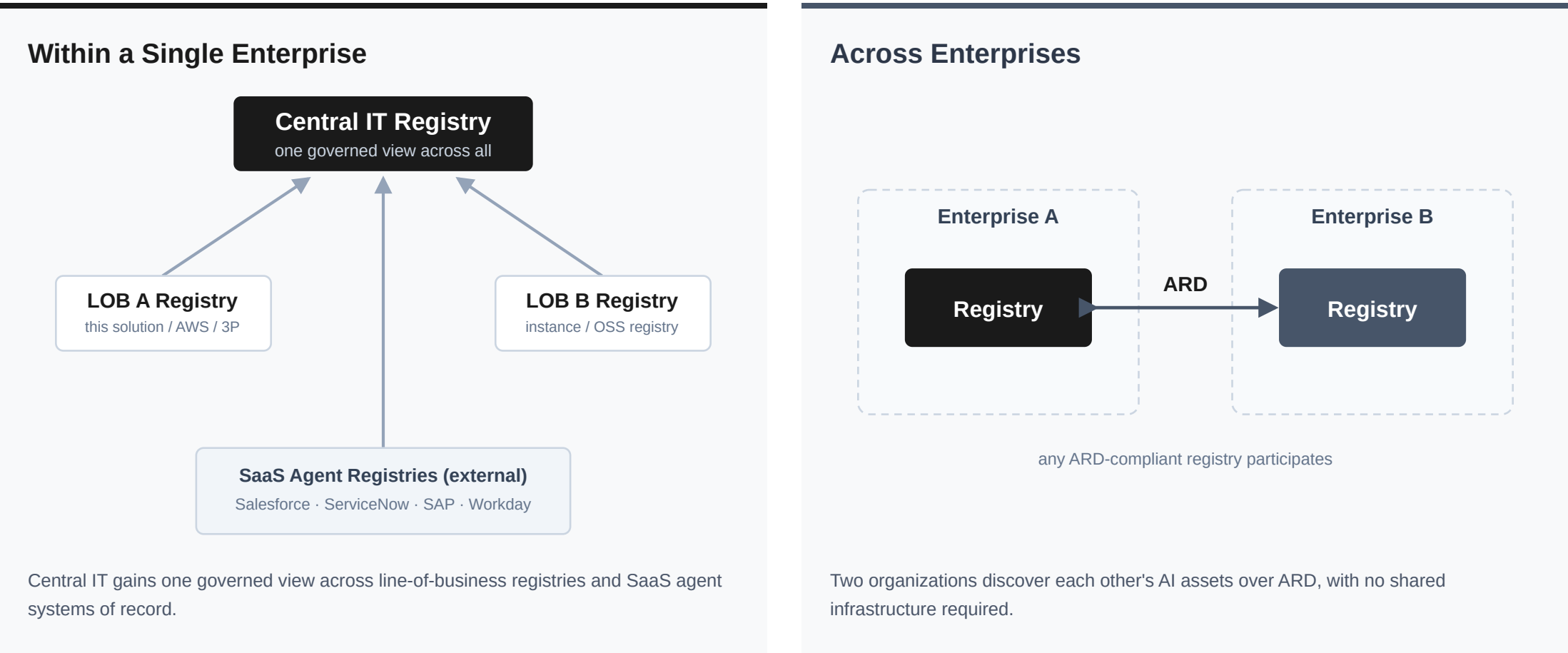
Verified agents display a trust badge with certificate details

IETF Internet-Draft (draft-narajala-ans-00) -- Powered by GoDaddy ANS, integrated into the open-source AI Registry

Federation: Two Scenarios

CONCEPT

Federation links registries so AI assets are discoverable across boundaries. It appears two ways: **within one enterprise** and **across enterprises**. Any participant can be an instance of this solution or any ARD-compliant registry.



Participants can be anything ARD-compliant: instances of this registry, AWS Agent Registry, Anthropic, Workday ASOR, or other third-party and open-source registries.

Agentic Resource Discovery (ARD)

Full **ARD v1.0** implementation -- vendor-neutral, standards-track discovery so any ARD-aware client, agent, or registry can find, search, and cross-reference assets.

Publisher

Anonymous, crawlable `/well-known/ai-catalog.json` of public + enabled assets, each a domain-anchored URN with the correct IANA media type.

Registry

Search & browse: `POST /api/ard/search` and `GET /api/ard/agents` -- JWT-required and access-scoped, wrapping the same semantic engine.

Federation

Ingest any ARD-conformant registry's `ai-catalog.json` into one unified index, with domain-anchored trust and an SSRF-guarded crawler.

```
{
  specVersion: "1.0",
  - host: {
    displayName: "AI Registry",
    - trustManifest: {
      identity: "https://mcpgateway.ddns.net",
      identityType: "https"
    }
  },
  - entries: [
    - {
      identifier: "urn:air:mcpgateway.ddns.net:server:realserverfaketools",
      displayName: "Real Server Fake Tools",
      type: "application/mcp-server-card+json",
      url: "https://mcpgateway.ddns.net/api/public/servers/realserverfaketools/server.json",
      description: "A collection of fake tools with interesting names that take different parameter types",
      - tags: [
        "demo",
        "fake",
        "tools",
        "testing"
      ],
      - capabilities: [
        "quantum_flux_analyzer",
        "neural_pattern_synthesizer",
        "hyper_dimensional_mapper",
        "temporal_anomaly_detector",
        "user_profile_analyzer",
        "synthetic_data_generator"
      ],
      - representativeQueries: [
        "demo tools",
        "fake tools",
        "tools tools"
      ],
      version: "v1.0.0",
      updatedAt: "2026-06-19T01:07:00.841437Z"
    },
    + { ... },
    + { ... },
    + { ... },
    + { ... },
  ]
}
```

Live `/well-known/ai-catalog.json`

A format adapter over existing registry data -- discover → resolve → connect, with no new storage.

External & Peer Registries

Two kinds of federation partner: **peer registries** you operate, and **external registries** you integrate.

Peer Registries

P2P Federation

Bidirectional sync between registry instances, with configurable modes: all, whitelist, or tag filter.

Registry Card

Standardized discovery via `/well-known/registry-card` advertising capabilities, auth endpoints, and contact info.

Peers are instances of this solution or any ARD-compliant registry.

External Registries

AWS Agent Registry

Federate MCP servers, A2A agents, and skills. Multiple registries across accounts and regions (cross-account via IAM role assumption), selective sync, and cascade cleanup.

Anthropic Registry

Import curated MCP servers from `registry.modelcontextprotocol.io`. Public API, no auth: set `ANTHROPIC_REGISTRY_ENABLED=true`.

Workday ASOR & SaaS

Federate Workday-managed agents (Agent System of Record) via OAuth, mapped to A2A agent cards. Salesforce, ServiceNow, and SAP plug in through the same adapter interface.

Unified governance: federated assets inherit the same scanning, RBAC, and audit as locally registered items. Enable each source with a single env var (for example `AWS_REGISTRY_FEDERATION_ENABLED=true`).

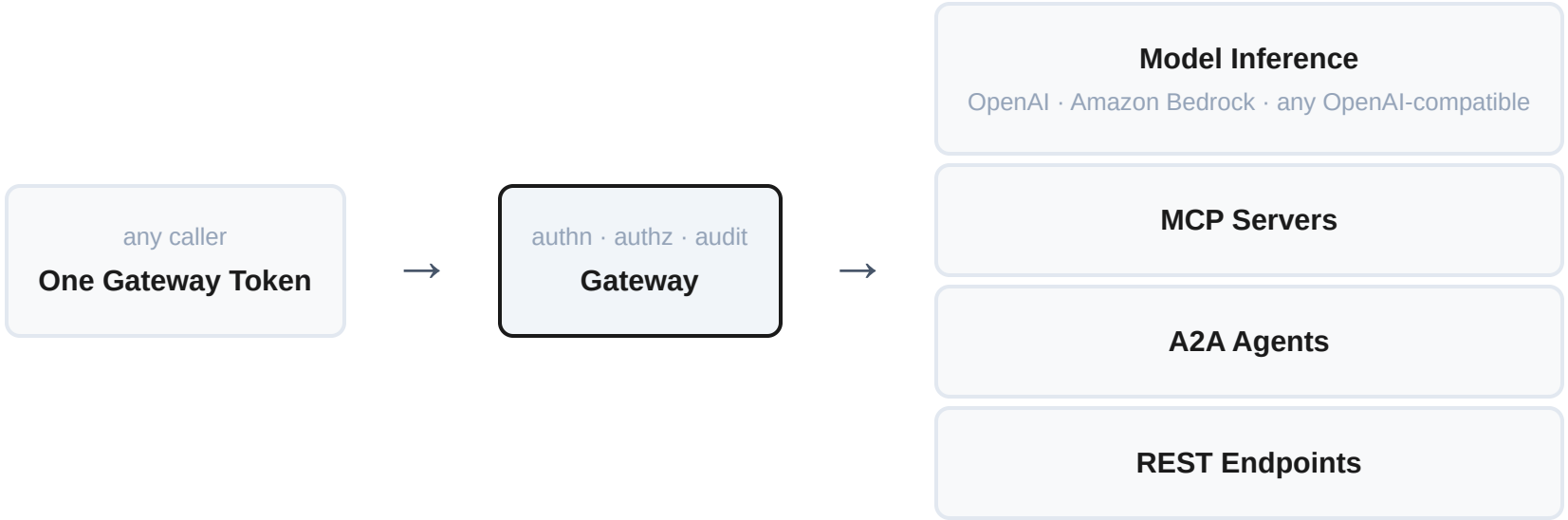
03

Gateway: Connect Securely

Every client connects once. The gateway carries each user's own identity to every backend — and no credential ever lives in agent code.

One Gateway for Any Resource

New in 1.30.0: the gateway fronts any HTTP backend, not only MCP servers. One token and one URL shape cover four kinds of traffic.



Keys never leave the registry

The backend URL and its credential stay in the registry, encrypted at rest and injected on egress, so no provider key lands on a laptop. Four credential models: keyless, operator key, caller passthrough, per-user vault.

Streaming, with ceilings

Per-entity response streaming carries SSE and Amazon Bedrock event-stream bodies incrementally, under bounded concurrency, byte, and duration limits.

The same governance

The same scopes, audit trail, and rate limits as the rest of the registry. A legacy methods: ["all"] grant authorizes no HTTP verb, so nothing gains access by accident.

Callers reach everything at `/gateway/{type}/{name}`. Off by default, per-entity opt-in, no schema migration.

Per-User Egress Auth: 3LO & OBO

SHIPPED · 1.27.0

The gateway brokers each user's own identity to third-party backends -- no shared service credentials, no tokens in agent code.

3LO -- Bring Your Own Account

Each user connects their own GitHub, Slack, Atlassian, or Google account once (three-legged OAuth). The gateway vaults the token in OpenBao / AWS Secrets Manager and injects it on egress.

OBO -- On-Behalf-Of Exchange

For same-trust-domain backends, the gateway performs RFC 8693 token exchange so the downstream server sees the real user identity. Target audiences are an audited, explicit allowlist.

Per-User, Auditable, Revocable

Tokens are scoped to the individual, masked in logs, and revoked by disconnecting the account. Direct PAT egress is next.

Per-User Egress Auth (OAuth)

Let each user connect their own third-party account (GitHub, Slack, ...). The gateway injects the user's token on egress. Leave provider blank to disable. Register this callback URL in your OAuth app:
<https://mcpgateway.ddns.net/oauth2/egress/callback>

Provider

Atlassian

Client ID

T9

Client Secret

leave blank to keep current

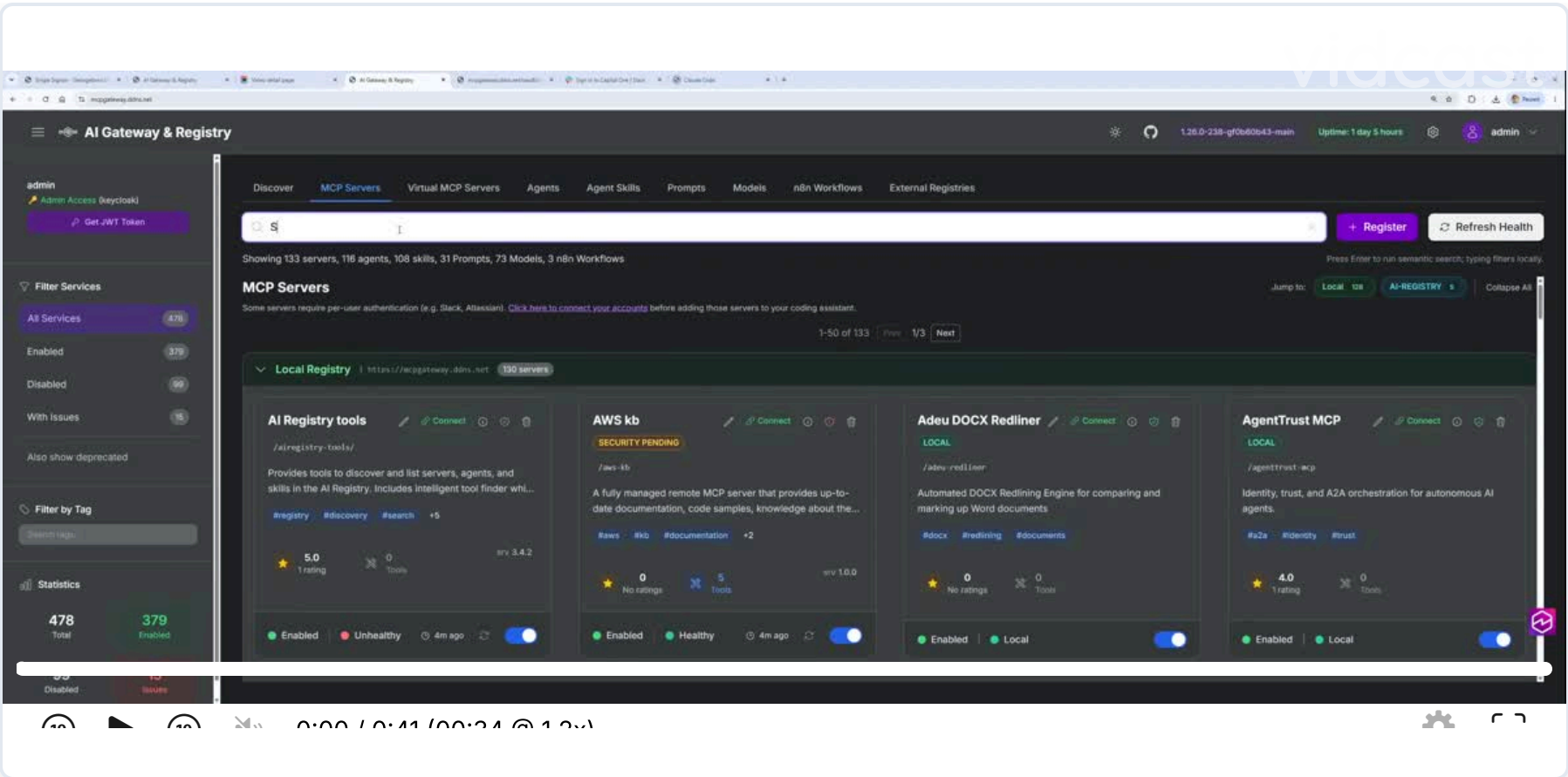
Scopes

read:jira-work, write:jira-work, read:jira-user, read:confluence-content.all, wr

Per-user egress auth config -- connect a provider, vault the token, inject on egress

Egress Auth in Action

A user connects their own account, then an agent calls the backend on their behalf -- token vaulted and injected by the gateway.



Video not playing? [Watch the egress auth demo](#)

One-Command OAuth for Coding Assistants

DCR (Dynamic Client Registration) and CIMD (Client ID Metadata Document) enable zero-config authentication. The assistant registers itself, the user clicks Allow, done.

DCR (RFC 7591)

Assistant self-registers with Keycloak via the gateway's published OAuth metadata.
No pre-provisioned client IDs needed. Per-user tokens with group-based entitlements.
Works today: Claude Code, Codex CLI, Claude.ai Connectors, Kiro

CIMD (draft-parecki)

For IdPs without DCR (Microsoft Entra ID): client publishes its metadata at a URL the IdP fetches. Same one-command UX, different wire protocol. Gateway acts as CIMD consumer.
Status: in development

▶ Claude Code Demo

▶ Codex CLI Demo

Umbrella Issue #988

One Front Door, Full Traffic Control

Because every call passes through one ingress, the gateway can shape, route, and protect all of it — MCP and agent-to-agent alike.

A2A Routing Inter-agent traffic optionally flows through the same ingress, so agent calls get the same fine-grained access control, audit, and identity brokering as MCP servers. Backends stay private.	Rate Limiting & Quarantine Identity-, group-, and target-aware request limits enforced at the auth hop — cap a runaway agent without throttling the team, or kill-switch any caller or target instantly.	Versions & Virtual Servers Multiple server versions behind one endpoint with instant rollback, plus virtual servers that aggregate tools from many backends under per-tool access control.
Upgrade a server, roll it back, cap a noisy caller — without touching a single client.		

A2A Reverse-Proxy Routing

SHIPPED · 1.27.0

Optionally route inter-agent (agent-to-agent) traffic through the gateway too, so agent calls get the same fine-grained access control we already apply to MCP servers.

One Endpoint for Agents

The gateway fronts your A2A agents the same way it fronts MCP servers -- callers hit a single ingress and the gateway brokers the rest. The agent's real backend stays private.

Per-Call FGAC

Each agent is gated per call with its own `invoke_agent` grant. Access decisions are enforced at the gateway, not left to the agent.

Same Governance as MCP

Unified auth, audit, and scoping across MCP servers and A2A agents -- one control plane for every hop of agent traffic.

Inter-agent calls now inherit the gateway's **fine-grained access control, audit, and identity brokering** -- opt-in, backends stay private.

04

Govern

Your IdP, your scopes, your audit trail — enforced on every call, with rate limits and quarantine when you need them.

Govern: Your IdP, Your Scopes, Your Audit Trail

No new identity silo. The platform plugs into the identity provider you already run and enforces the policies you already have.

Multi-Provider IAM Six IdPs -- Keycloak, Microsoft Entra ID, Okta, Auth0, AWS Cognito, and PingFederate -- with a harmonized management API.	Fine-Grained Access Scopes define which servers, methods, tools, and agents each user or service account can access.	Rate Limits & Quarantine Identity-, group-, and target-aware request limits, plus an admin kill switch for any caller or target.
Audit Logging Comprehensive audit trails with credential masking, TTL retention, and SOC 2/GDPR compliance.	M2M Auth OAuth2 Client Credentials flow for AI agent identity with service accounts.	Static Token Auth API key access for CI/CD pipelines and trusted network environments without IdP setup.

Application-Level Rate Limiting

SHIPPED · 1.27.0

The gateway now does AI traffic management -- identity-, group-, and target-aware request limits enforced at the auth-server hop, complementary to coarse per-IP edge limiting.

Cap the Caller

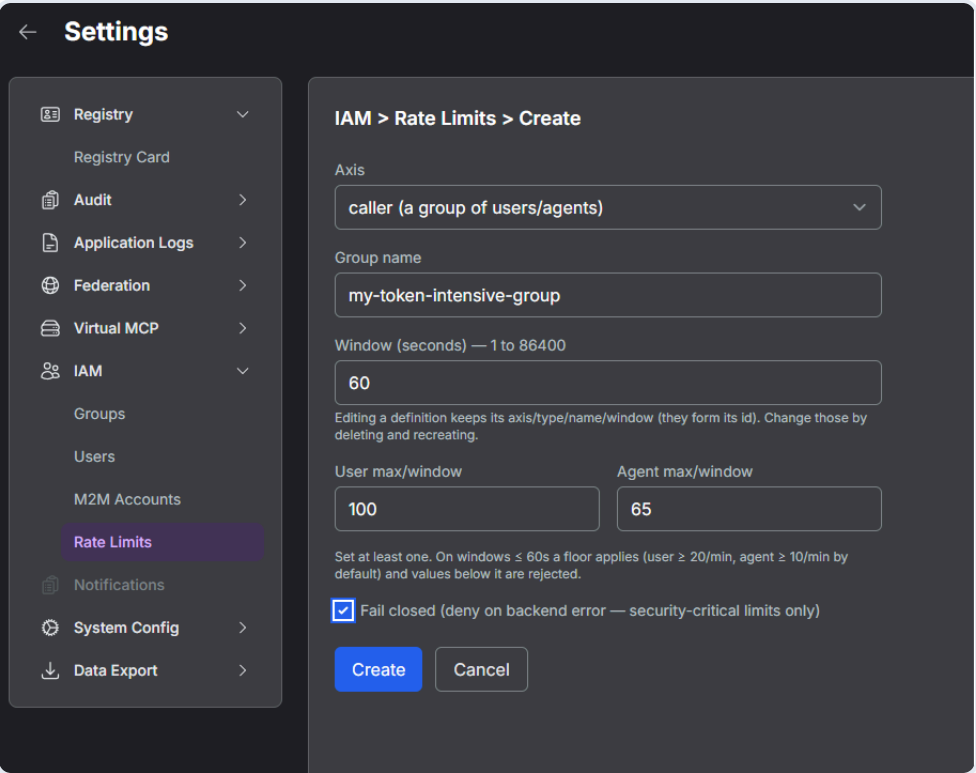
Limit a group of users or agents (M2M) per window. Separate user and agent ceilings, with lockout-safeguard floors so you can never accidentally throttle a group to zero.

Protect the Target

Cap requests to a busy MCP server or A2A agent regardless of who is calling, so one hot backend can't be overwhelmed.

Fail-Open or Fail-Closed

Global fail-open by default so a counter-store blip never becomes a gateway outage; flip security-critical limits to fail-closed per definition. Managed at runtime via API, CLI, or the UI.



Settings → IAM → Rate Limits -- per-group user and agent ceilings

05

Run It: Operations, Proof & Roadmap

Deploy anywhere, observe everything — and the momentum behind the project.

Runs Where You Run

Deploy Anywhere

Amazon EKS (Helm), ECS Fargate (Terraform), EC2 with Docker Compose, or rootless Podman on a laptop. Helm charts work on any Kubernetes: EKS, AKS, GKE, or your own data center.

Observe Everything

Native OpenTelemetry metrics with Grafana dashboards, an always-on Prometheus endpoint, and direct OTLP push to Datadog, New Relic, Honeycomb, or Grafana Cloud — no collector needed.

Operate Simply

Conversational registry CLI, visual IAM settings UI, RESTful management API with Python client, and config export as ENV, JSON, TFVARS, or YAML for automation.

4

6

6,400+

Deployment TargetsIdentity ProvidersTest Functions in CI

Same configuration surface across all targets — a platform team runs it **the way they run everything else**.

Community Adoption

A community-driven project with growing momentum since June 2025

920
GitHub Stars

62
Contributors

237
Forks

2,245
Lifetime Deployments



956
Merged PRs

531
Issues Closed

35
Releases

26.7%
Customers running 3+ days

ECS ≈ K8s ≈ Docker
among active customers

Listed on GoodAIIist — curated directory of top AI repositories

Featured Coverage

AWS MACHINE LEARNING BLOG

MAY 2026

Securing AI agents: How AWS and Cisco AI Defense scale MCP and A2A deployments

A joint AWS and Cisco perspective on how the MCP Gateway & Registry, with Cisco AI Defense scanners, secures the supply chain for MCP servers and A2A agents at enterprise scale -- registration-time scanning, runtime trust, and federation across AWS Agent Registry, Anthropic, and Workday ASOR.

[Read on aws.amazon.com](#) →

AWS OPEN SOURCE BLOG

JUNE 2026

Governing AI assets at scale with MCP Gateway and Registry

How the MCP Gateway & Registry gives enterprises one solution to discover, govern, and securely connect MCP servers, A2A agents, and skills -- unified semantic discovery, OAuth2 across IdPs, visibility controls, and federation, all open source.

[Read on aws.amazon.com](#) →

Roadmap & Shipping Velocity

35 releases since June 2025, built in the open — latest: 1.30.0, August 2026. Three themes define the recent arc; the full release timeline lives in the release notes on GitHub.

Per-User Egress Identity SHIPPED 3LO + OBO identity brokering with a vaulted credential store and fail-closed egress guards — now with public-client PKCE and multi-IdP login brokering.	AI Traffic Management SHIPPED A2A reverse-proxy routing with per-call access control, plus identity- and target-aware rate limiting with quarantine.	Any-Asset Registry SHIPPED Custom entity types, hybrid semantic search, ARD v1.0 federation, and leaner read APIs via metadata projection (1.30.0).
Next: Any-Asset Routing IN PROGRESS · #1565 Today the gateway routes MCP servers and A2A agents. Next: securely proxy any registered resource — skills, agents, custom entities — through the same governed front door, with central auth, audit, and SSRF defenses designed in from the start. The foundation PR is already open — the step that makes this a true AI gateway.	Next: Frictionless Auth Everywhere Client ID Metadata Documents and RFC 8693 token exchange, so coding assistants and brokered MCP servers connect with least friction across every IdP — already 75% landed.	Next: Registry Copilot An embedded chat and agent-builder experience for discovering assets and composing agents from inside the registry, plus one-command export to AWS Agent Registry.

We build in the open — tell us what you need →

06

Get Started

Try it, connect a coding assistant, and get in touch.

DEMO & LINKS



Securing AI agents: How AWS and Cisco AI
Defense scale MCP and A2A deployments

[AWS Machine Learning Blog -- joint AWS + Cisco perspective](#)



AWS Show & Tell

[YouTube -- MCP Gateway & Registry deep dive](#)



Claude Code + DCR Demo

[Vidcast -- One-command OAuth integration walkthrough](#)



Codex CLI + DCR Demo

[Vidcast -- OpenAI Codex connecting via DCR OAuth](#)



Platform Demo

[Vidcast -- Hands-on walkthrough of the platform](#)



Roadmap

One AI Registry. Unified Control.

3

AWS Targets

ECS Fargate, EKS, EC2

6

Identity Providers

Keycloak, Entra ID, Okta, Auth0, Cognito,
PingFederate

Open

Source

Apache 2.0 License



github.com/agentive-community/mcp-gateway-registry

Scan to get started · Open source, Apache 2.0 License

Quick Start

Documentation

AWS Deployment

AWS Workshop

Demo Video

Community